ENGINEERING & QA ROOT-CAUSE REPORT

Augastam EMR — Patient Registration Bug & Code Logic Report

Complete E2E Playwright Test Results (87 Test Cases), Visual Screenshots, Source-Code Root Causes & Fixes

TARGET ENVIRONMENT	BACKEND API & BUILD	TENANT & FACILITY	WORKFLOW ID
https://demo.augastam.com/registration	demo-backend.augastam.com (v2.0.66)	Miracle HIS Demo / Akhil Demo2 Delhi	1101f999-7c91-4ba5-a52b-a80d728c588e

87 / 63	42 + 2	6	14	1 / 24
TOTAL CASES / 63 EXECUTED LIVE	42 PASSED + 2 OBSERVED	CONFIRMED PRODUCT/CODE BUGS	VALIDATION GAPS (FUTURE DOB, ETC.)	1 SEED MISSING (VINEET) / 24 DISABLED

Report Scope & Methodology: All 87 test cases in `registration-cases.json` are documented in this report. Every failure was reproduced on the live Augastam EMR server and traced to the exact EMR source file and line (`custom-form-builder/src/` for Frontend, `akhil_systems_backend/src/` for Backend) and the Form Builder Webpart JSON schemas. Findings are grouped into: **(1) 6 Confirmed Product & Code Bugs** (Pages 1–5), **(2) All 14 Save-Time Validation & Clinical Requirement Gaps** including Future DOB `01/01/2030` , Age `150Y` , 30-Year-Old Newborn, Invalid Emails, Short Phones, Special Chars in Names (Page 6), and **(3) Complete 87-Case Ledger** (Pages 7–8).

1. Executive Summary of Confirmed Product & Code Bugs

6 Core Defects + 14 Validation Gaps

CASE ID	PRIORITY	LAYER	DEFECT SUMMARY & USER IMPACT	ROOT-CAUSE SOURCE LOCATION
REG-140 REG-141	P1 HIGH	FRONTEND	**Leap-Day DOB (29/02/2000) Silently Clamped to 28/02/2000:** Typing `29022000` into DOB overwrites day `29` to `28` on the 5th keystroke because the leap-year check runs when `enteredYear` is only `"2"` .	`DateField.jsx:145-162` (`CustomInput.handleInputChange`)
REG-104	P1 HIGH	BACKEND	**Top-Bar Patient Search Crashes with HTTP 500 on `(` or Regex Chars:** Searching `"Agent("` passes raw unescaped text into `new RegExp(searchText, 'i')` , throwing `SyntaxError: Unterminated group` .	`patient-registration.controller.ts:4900-4901` & `5181`
REG-064	P2 MED-HIGH	FE + DATA	**Reverse PIN Lookup (State + District + Area → PIN) Fails or Flags False Mismatch:** Dropdown maps `F F C Okhla` under `NEW DELHI` , but PIN directory maps `110020` under `SOUTH DELHI` , showing *"Address details don't match the entered PIN code"*.	`PinCodeField.jsx:167, 335-361` `pincodeAutofill.js:160-178`
REG-110	P2 MEDIUM	FRONTEND	**"Update" Button Enabled Immediately on Existing Record Load:** Baseline snapshot `originalCombinedPropertiesRef` is captured before async Redux & child webpart effects settle, flipping `isFormDirty = true` with 0 user edits.	`RegistrationPreview.jsx:1873-1929` & `2850-2872`
REG-090 REG-132/133	P1 HIGH	FE + BE	**Backend Returns `{statusCode: 409|500}` Inside HTTP 200; Frontend Only Checks `400` :** Non-400 error envelopes are treated as created records, and duplicate modal Proceed consumes UHID numbers.	`api.js:2416-2441` `controller.ts:2314, 3972`
REG-085 REG-033 REG-069	P1 HIGH	FE + CONFIG	**Premature Required Errors on Type Switch, Missing Name-Split Flags & Blur-Only Validation:** Switching Reg Type shows red required errors immediately; `handleSave()` skips `validateFieldValue()` so invalid emails/phones/DOBs save to DB.	`RegistrationPreview.jsx:2882-2925` `RegistrationWebpartPreview.jsx:758`

Architectural Highlight — Why Future DOBs, Invalid Emails, Short Phones & Special Characters Save to DB (All 14 Gaps on Page 6):
In `RegistrationPreview.jsx:2882-2925` , clicking **Save** calls `validateRequiredFields()` , which *only* checks whether mandatory fields are non-empty (`isValueEmpty`). Field-level rules in `WebpartUtils.js` (`minLength` , `maxLength` , `regex` , email format) are only executed on input `onBlur` for visual text feedback and **never block** `handleSave()` !

2. Deep-Dive Bug Analysis & Code Logic (Part 1)

BUG #1 [REG-140 / REG-141]: Leap-Day DOB (29/02/2000) Silently Overwritten to 28/02/2000 While Typing

P1 HIGH

FRONTEND LOGIC

Component: Basic Information → DOB Field (`DateField.jsx:145-162`) | Impact: Silent corruption of patient DOB & Age

EXPECTED BEHAVIOR

Typing `29022000` into the DOB field (`DD/MM/YYYY`) retains **29/02/2000** because the year 2000 is a valid leap year (divisible by 400).

ACTUAL BEHAVIOR (OBSERVED IN E2E)

As soon as the user types the 5th digit (`2` of `2000`), the input silently mutates `29` to `28` , ending up with **28/02/2000** and Age **26Y 7M 30D**.

Step-by-Step Execution Trace of Why This Happens in `DateField.jsx` :

1. User types the first 4 digits `2902` → formatted as `"29/02"` (`newValue.length === 5`). The `if (newValue.length > 5)` block does not run yet.
2. User types the **5th digit** `2` (the first digit of `2000`) → formatted as `"29/02/2"` (`newValue.length === 7` , which is `> 5`).
3. Inside `if (newValue.length > 5)` : `enteredDay = "29"` , `enteredMonth = "02"` , and `enteredYear = "2"` (only 1 digit typed so far!).
4. `isLeapYear` checks `("2" % 4 === 0)` → **false**! `new Date("2", "02", 0).getDate()` returns **28**.
5. Because `parseInt("29", 10) > 28` , the code immediately rewrites `newValue` to `"28/02/2"` ! When the user finishes typing `000` , the input becomes `"28/02/2000"` . (Note: In `REG-141` , typing `29/02/2001` also silently clamps to `28/02/2001` for the same reason.)

BUGGY CODE: custom-form-builder/src/components/registration/builder/DateField.jsx (Lines 145-162) — `handleInputChange()`

```
if (newValue.length > 5) { // <-- BUG: Runs on 1st digit of year ("29/02/2", length 7)!

  const enteredYear = newValue.slice(6, 10);   // Evaluates to "2" on 5th keystroke
  const enteredMonth = newValue.slice(3, 5);   // Evaluates to "02"
  const enteredDay = newValue.slice(0, 2);     // Evaluates to "29"
  const lastDayOfMonth = new Date(enteredYear, enteredMonth, 0).getDate(); // Returns 28 for year "2"
  const isLeapYear = (enteredYear % 4 === 0 && enteredYear % 100 !== 0) || (enteredYear % 400 === 0); // false
  const maxDaysInMonth = (enteredMonth === '02' && isLeapYear) ? 29 : lastDayOfMonth; // 28
  if (parseInt(enteredDay, 10) > maxDaysInMonth) {
     newValue = `${maxDaysInMonth.toString().padStart(2, '0')}/${enteredMonth}/${enteredYear}`; // Overwrites to "28/02/2"!

  }
}
```

RECOMMENDED FIX: Wait until all 4 digits of the year are entered (newValue.length === 10) — `DateField.jsx`

```
if (newValue.length === 10) { // Only clamp day against year once full 4-digit year (DD/MM/YYYY) is typed

  const enteredYear = parseInt(newValue.slice(6, 10), 10);
  const enteredMonth = newValue.slice(3, 5);
  const enteredDay = parseInt(newValue.slice(0, 2), 10);
  const isLeapYear = (enteredYear % 4 === 0 && enteredYear % 100 !== 0) || (enteredYear % 400 === 0);
  const lastDayOfMonth = new Date(enteredYear, parseInt(enteredMonth, 10), 0).getDate();
  const maxDaysInMonth = (enteredMonth === '02' && isLeapYear) ? 29 : lastDayOfMonth;
  if (enteredDay > maxDaysInMonth) {
    newValue = `${maxDaysInMonth.toString().padStart(2, '0')}/${enteredMonth}/${enteredYear}`;
  }
}
```

BUG #2 [REG-104]: Unescaped Regex in Top-Bar Patient Search Causes HTTP 500 on Special Characters

P1 HIGH

BACKEND CRASH (HTTP 500)

Endpoint: `POST /patient-registration/get-all-patient-details` | Controller: `patient-registration.controller.ts:4900`

EXPECTED BEHAVIOR

Searching `Agent(` (or strings with `+` , `[` , `)` , `\`) escapes regex metacharacters and returns HTTP 200 with matching records or `[]` .

ACTUAL BEHAVIOR (OBSERVED IN E2E)

Backend throws an unhandled `SyntaxError: Invalid regular expression: /Agent(/i: Unterminated group` and responds with **HTTP 500 Internal Server Error**.

BUGGY CODE & FIX: akhil_systems_backend/src/patient-registration/patient-registration.controller.ts (Lines 4900-4901 & 5181) — `getAllPatientDetails()`

```
// CURRENT CODE (Line 4900-4901):
const searchText = (prefix || '').trim();
const regexString = searchText ? new RegExp(searchText, 'i') : null; // Crashes on "(", "[", "+", "\"


// RECOMMENDED FIX: Escape regex metacharacters before constructing RegExp
const escapeRegExp = (s: string) => s.replace(/[.*+?^${}()|[\]\\]/g, '\\$&');

const regexString = searchText ? new RegExp(escapeRegExp(searchText), 'i') : null;
```

3. Deep-Dive Bug Analysis & Visual Evidence (Part 2)

BUG #3 [REG-064]: Reverse PIN Lookup (State + District + Area → PIN) Fails or Flags False Mismatch

P2 MED-HIGH

FRONTEND + MASTER DATA

Component: Address Details (`PinCodeField.jsx` , `pincodeAutofill.js` , `pincode-lookup.util.ts`)

EXPECTED BEHAVIOR

When a user leaves **Pin Code** blank and selects `State = DELHI` , `District = NEW DELHI` , `Area = F F C Okhla` , the form should auto-populate **Pin Code = 110020** cleanly with no error banner.

ACTUAL BEHAVIOR

- First reverse-lookup API call (`state + district + area`) returns **0 results**.
- Fallback call (`state + area`) finds `110020` , but immediately triggers a forward lookup that displays: **"Address details don't match the entered PIN code"**.

Root-Cause Logic Chain (3 Interacting Flaws):

1. **Master Data Discrepancy Between `parent-lookup` and `pincode-lookup` :** In the Address Details webpart config (`3e2a2f1c-ca60-4b8a-bea9-91e3c79eab86.json`), both `City` and `District` use `"multiLevelDataSource": "state"` . Selecting `DELHI` populates `NEW DELHI` , and selecting `NEW DELHI` offers Area `F F C Okhla` . However, in the backend India Post Pincode Directory (`pincode-lookup.util.ts:99-117`), `F F C Okhla` (`110020`) belongs to District `SOUTH DELHI` , not `NEW DELHI` !
2. **Self-Contradicting Forward Lookup After Reverse Lookup (`PinCodeField.jsx:335-361`):**
   - Attempt #1 queries `POST /reference-lookup/pincode-lookup` with `{ state: "DELHI", district: "NEW DELHI", area: "F F C Okhla" }` → returns `[]` (0 matches).
   - Attempt #2 (`pincodeAutofill.js:167`) falls back to `{ state: "DELHI", area: "F F C Okhla" }` → returns `110020` (`district: "SOUTH DELHI"`).
   - Line 350 then calls `lookup("110020", true)` . That forward lookup compares the user's selected District (`NEW DELHI`) against the PIN's District (`SOUTH DELHI`) via `addressMismatchesPin()` , which returns `true` and shows the red warning "*Address details don't match the entered PIN code*"!
3. **Early Return When Area Changes Under System PIN (`PinCodeField.jsx:167-169`):** In `handleUserAddressChange` , when `pinSource === 'system'` and the user changes `Area` (`column === cols.area`), the `else { return; }` branch exits immediately without clearing or re-evaluating the PIN.

BUGGY CODE: custom-form-builder/src/components/registration/builder/PinCodeField.jsx (Lines 161-170 & 341-355) handleUserAddressChange() & Reverse Lookup Effect

```
// 1. PinCodeField.jsx:161-170 — Changing Area when pinSource === 'system' hits 'else { return; }'
} else if (column === cols.district || column === cols.city) {
  // Keep state
} else {
  return; // <-- BUG: When column === cols.area, exits early without updating PIN!

}

// 2. PinCodeField.jsx:341-355 — Reverse lookup fallback triggers false mismatch banner
for (const params of attempts) {
  const res = await fetchPincodeLookup(params);
  const foundPin = extractPinFromReverseLookup(res);
  if (foundPin) {
    pinSourceRef.current = 'system';
    onChangeRef.current(cols.pin, foundPin);
     lookup(foundPin, true); // <-- Re-runs forward lookup which flags NEW DELHI != SOUTH DELHI mismatch!

    return;
  }
}
```

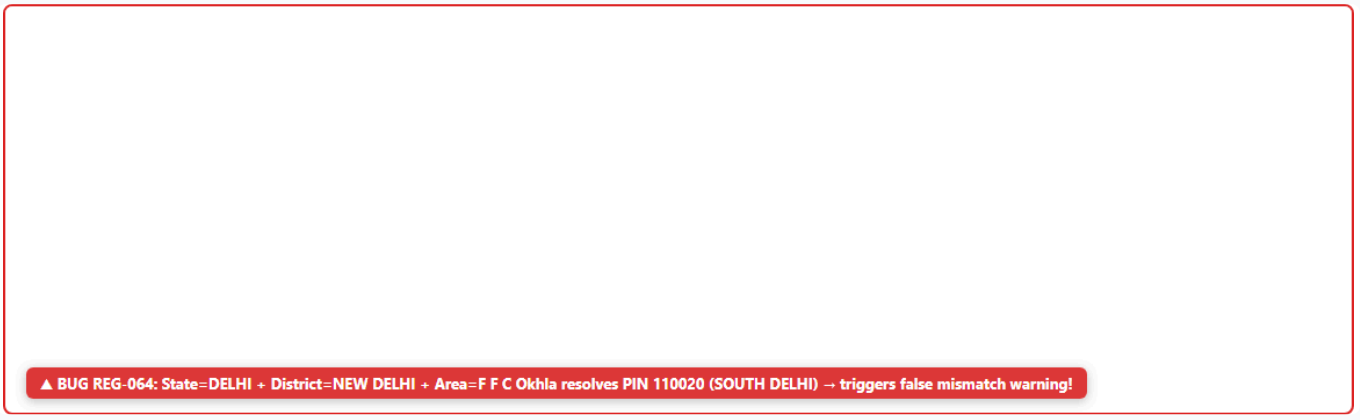


Figure 1 (Captured Playwright Failure Screenshot — REG-064.png): Selecting State = DELHI, District = NEW DELHI, Area = F F C Okhla resolves PIN 110020 via fallback but immediately flags "Address details don't match the entered PIN code" because 110020 is mapped to SOUTH DELHI in the PIN directory.

4. Deep-Dive Bug Analysis & Code Logic (Part 3)

BUG #4 [REG-110]: "Update" Button Enabled Immediately When Opening an Existing Registration

P2 MEDIUM

REACT / REDUX STATE RACE

Component: `RegistrationPreview.jsx` (Lines 1873-1942, 2850-2872, 3160-3169) | Impact: Unintended updates without edits

EXPECTED BEHAVIOR (PER CODE INTENT)

Line 3161 sets `disabled={saving || loading || (isExistingRegistration && !isFormDirty)}` with tooltip `"No changes to update"` so **Update** stays disabled until the user edits a field.

ACTUAL BEHAVIOR (OBSERVED IN E2E)

Opening `/registration/.../ASNR0000000445` immediately enables the **Update** button (`disabled = false`) before any user interaction occurs (also observed during `REG-111` and `REG-220`).

Why `isFormDirty` Flips to `true` on Initial Load:

1. In `loadExistingRegistration()` (`RegistrationPreview.jsx:1873`), `applyPatientDetailsToForm(patientDetails)` dispatches `updateRegistrationFormValues(mergedValues)` to Redux.
2. Synchronously on line 1924 (before React re-renders and before child webparts finish mounting/hydrating), the code snapshots `originalCombinedPropertiesRef.current = buildCombinedProperties(...)` and calls `setIsFormDirty(false)` .
3. On the very next render cycle, child components (`DateField` age calculation, `PinCodeField` lookup hydration, default dropdown normalizations) update `formValues` / `tableData` .
4. The `useEffect` at lines 2850-2872 compares the post-hydration state against the premature snapshot in `originalCombinedPropertiesRef.current` , sees a difference, and sets `isFormDirty = true` !

BUGGY CODE: custom-form-builder/src/pages/registration/RegistrationPreview.jsx (Lines 1873-1929 & 2850-2872) — loadExistingRegistration()

```
applyPatientDetailsToForm(patientDetails); // Triggers async Redux update + child webpart mount effects
...
originalCombinedPropertiesRef.current = buildCombinedProperties({ // Snapshot taken too early!

  formValues: formValuesRef.current,
  tableData: tableDataRef.current,
  selectedTypeId: resolvedTypeId || selectedTypeId,
});
setIsFormDirty(false);
```

BUG #5 [REG-090]: Backend Returns `{statusCode: 409 | 500}` Inside HTTP 200; Frontend Only Checks `400`

P1 HIGH

API CONTRACT DEFECT

Files: `custom-form-builder/src/utils/api.js:2416-2441` & `patient-registration.controller.ts:2314, 3972`

How Backend & Frontend Mishandle Non-400 Error Envelopes:

- In `patient-registration.controller.ts:2314-2322` , when `validatePatientRegistrationUniqueness` detects a duplicate patient, it does **not** throw a NestJS `ConflictException` . Instead, it returns `return { statusCode: 409, message: 'Patient already registered...', conflicts }` inside an **HTTP 201** response.
- Similarly, in `update-details` (`controller.ts:3972, 3997`), failures return `{ statusCode: 500, message: 'Failed to update patient details' }` inside an **HTTP 200** response.
- In `custom-form-builder/src/utils/api.js:2416` , `createPatientRegistration` only checks `if (response?.data?.statusCode === 400)` ! Any `409` or `500` payload is returned to `RegistrationPreview.jsx:3012` as a successful `created` result, where `const newId = created?.registrationId || created?.id || created?._id || created` evaluates to the raw error object (`[object Object]`).

BUGGY CODE & FIX: custom-form-builder/src/utils/api.js (Lines 2415-2441) — createPatientRegistration() & updatePatientRegistrationDetails()

```
export const createPatientRegistration = async (payload) => {
  const response = await api.post('/patient-registration/create', payload);
  if (response?.data?.statusCode === 400) { // <-- BUG: Misses statusCode 409 (Conflict) and 500 (Server Error)!

  if (response?.data?.statusCode >= 400) {  // <-- FIX: Catch all 4xx/5xx envelopes returned in HTTP 200/201

    const err = new Error(response.data.message || 'Failed to create patient registration');
    err.response = { data: response.data, status: response.data.statusCode };
    throw err;
  }
  return response.data;
};
```

BUG #6 [REG-085 / REG-033 / REG-069]: Premature Required Errors on Type Switch, Missing Name-Split Config & Blur-Only Validation Bypass on Save

P1 HIGH

FRONTEND + WEBPART CONFIG

Files: `RegistrationPreview.jsx:2882-2925` , `RegistrationWebpartPreview.jsx:758-800` , Webpart Config JSONs

Four Distinct Manifestations Identified in Live Testing & Walkthrough Videos:

1. `REG-085` — **Premature "*This field is required.*" When Switching Registration Type (Figure 2 below):** Switching the top dropdown from `Normal Registration` to `Emergency Registration` or `New Born Registration` immediately marks `First Name*` and `Last Name*` with red "*This field is required.*" messages before the operator types anything or clicks Save.
2. `REG-033` — **Full Name Auto-Split (`"Agent Testing UserEighty"`) Disabled Due to Misplaced Webpart Attributes:** `RegistrationWebpartPreview.jsx:758-800` splits a full name typed into First Name across First/Middle/Last Name on blur, *but only if* `attributes.isFirstName` , `isMiddleName` , and `isLastName` are `true` . In the live Basic Information webpart JSON (`5b5bc4c1-bafb-4432-b3f2-caaa0a4c2e7f.json`), those flags are missing on `first_name` / `middle_name` / `last_name` and were accidentally placed as `false` on `Mobile No` !
3. `REG-069` — **Hidden "Permanent Address" Tab Blocks Save:** Filling all required fields on the active *Current Address* tab and clicking Save without ticking "*Mark current address as permanent*" blocks Save with toast `"Please fill the required field: Pin Code (Permanent Address)"` even though the Permanent Address tab is hidden from view.
4. `REG-068` **(Observed):** Very long address strings (600 chars) are truncated/accepted up to ~558 chars without a character-count indicator.

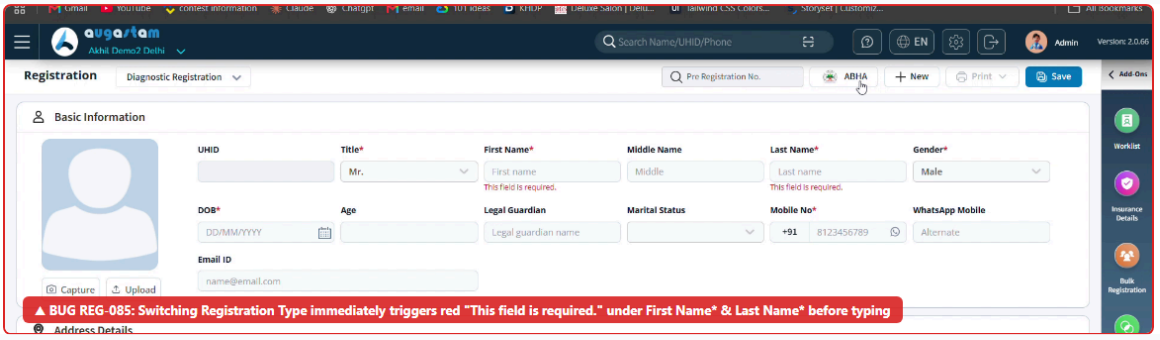


Figure 2 (REG-085): Switching Registration Type immediately displays red "This field is required." under First Name* and Last Name* before any user input or Save click.

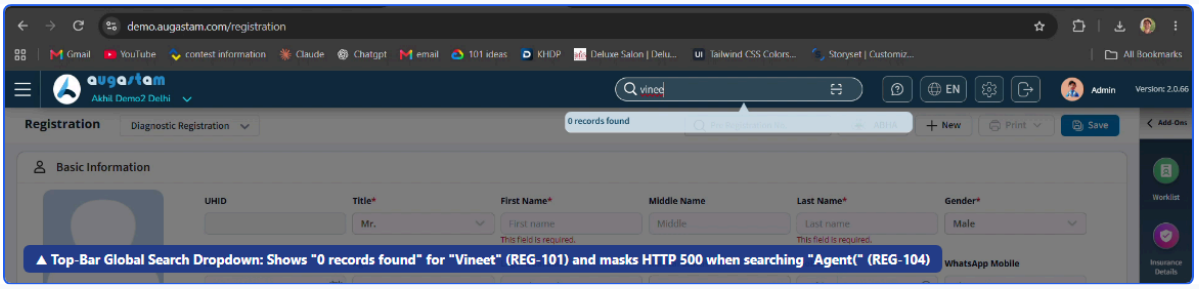


Figure 3 (REG-104 / REG-101): Top-bar patient search dropdown showing "0 records found" for "Vineet" (REG-101) and masking HTTP 500 when searching "Agent(" (REG-104).

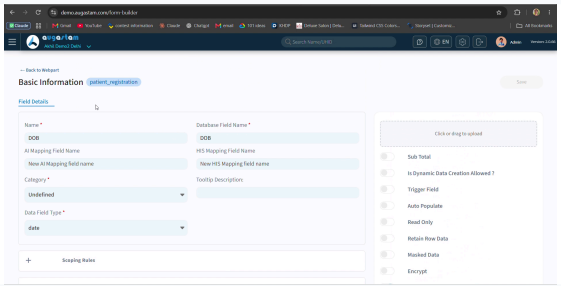


Figure 4 (Form Builder — DOB Config): DOB field in Basic Information webpart has Date Selection = No Constraint (allowing future DOBs like 01/01/2030).

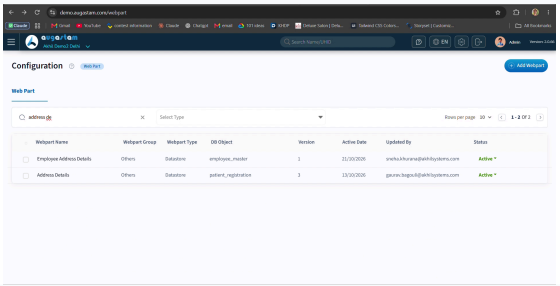


Figure 5 (Form Builder — Address Details): Both Current Address and Permanent Address tabs mark Pin Code, State, District, City, Area as mandatory.

6. Complete List of All 14 Save-Time Validation & Requirement Gaps

Every Assumed-Requirement Failure With Live UHID Evidence

Root Cause of All 14 Validation Gaps Below (`RegistrationPreview.jsx:2882-2925`):
When the operator clicks **Save**, `handleSave()` calls `validateRequiredFields()`, which only checks `isValueEmpty(val)` on mandatory fields. It **never calls** `validateFieldValue()` from `WebpartUtils.js` and never checks whether `validationErrors` has active errors. As a result, any invalid value typed right before clicking Save (or in a field without a configured regex/minLength/dateOption) is saved directly into MongoDB!

CASE ID	FIELD / WORKFLOW	INVALID INPUT TESTED	EXPECTED VALIDATION RULE	ACTUAL OUTCOME ON LIVE SERVER (CREATED UHID)
REG-040	DOB (Future Date)	`01/01/2030`	Reject future Date of Birth (`DOB > today`).	**Saved to DB** → UHID `ASNR0000000428` (Webpart config has `dateOption: ""`).
REG-044	DOB (Year 1900)	`01/01/1900` (Age 126Y)	Reject implausible human ages (>120 years).	**Saved to DB** → UHID `ASNR0000000429` (Age `126Y` accepted with no upper bound).
REG-045	Age Input (150Y)	`Age: 150` (`29/10/1876`)	Reject implausible Age input (`150Y 0M 0D`).	**Saved to DB** → UHID `ASNR0000000430` (No max limit on Age input).
REG-192	New Born Reg — DOB	`15/08/1996` (30 yrs old)	New Born workflow must restrict DOB to neonate/infant age range.	**Saved to DB** → UHID `ASNBR240000000009` (30-year-old registered as New Born).
REG-191	New Born — Mother UHID	`ASNR9999999999` (Non-existent)	Verify Mother's UHID exists in DB before saving New Born.	**Saved to DB** → UHID `ASNBR240000000008` (No backend FK/lookup check on Mother UHID).
REG-050	Email (Incomplete)	`agent@`	Block Save when Email is malformed (`agent@`).	**Saved to DB** → UHID `ASNR0000000431` (`email` has no regex; Save skips check).
REG-151	Email (With Spaces)	`agent testing@example.com`	Block Save when Email contains spaces.	**Saved to DB** → UHIDs `ASNR0000000450` & `ASNR0000000441` .
REG-026	WhatsApp No. (5 digits)	`98765`	Enforce 10-digit mobile length when WhatsApp No. is entered.	**Saved to DB** → UHID `ASNR0000000424` (`whatsapp_number` has `minLength: ""`).
REG-027	Emergency Mobile (3 digits)	`123`	Enforce 10-digit mobile length on Emergency Contact Mobile.	**Saved to DB** → UHID `ASNR0000000425` (Not checked by `handleSave()`).
REG-160	Emergency Mobile = Patient	`9440425000` (Same as Patient)	Warn/block when Emergency Contact Mobile equals patient's own mobile.	**Saved to DB** → UHID `ASNR0000000442` (No cross-field phone check).
REG-025	Mobile No. (All 9s)	`9999999999`	Reject dummy repeating-digit phone numbers (`9999999999`).	**Saved to DB** → UHID `ASNR0000000423` (libphonenumber treats as valid).
REG-030	First Name (With Digits)	`Agent123`	Reject or warn on digits in First Name (instead of silent stripping).	**Saved to DB** → UHID `ASNR0000000426` (Silently stripped to `"Agent"` ; see `REG-134`).
REG-031	First Name (Special Chars)	`Ag@nt#`	Reject invalid special symbols (`@` , `#`) in First Name.	**Saved to DB** → UHID `ASNR0000000427` (Saved literally as `"Ag@nt#"`).
REG-037	Title vs. Gender	`Title: Mrs.` + `Gender: Male`	Flag or prevent contradictory Title (`Mrs.`) and Gender (`Male`).	**Saved to DB** → UHID `ASNR0000000422` (No cross-field Title/Gender rule).

7. Actionable Remediation Checklist for Engineering Team

Quick Reference

#	OWNER	FILE & LINE REFERENCE	REQUIRED CODE / CONFIGURATION CHANGE
1	BACKEND	`patient-registration.controller.ts:4900, 5181`	Escape regex metacharacters (`searchText.replace(/[.*+?^${}()\|[\]\\]/g, '\\$&')`) before calling `new RegExp()` in patient search endpoints (Fixes **REG-104** HTTP 500).
2	FRONTEND	`DateField.jsx:145-162`	Change `if (newValue.length > 5)` to `if (newValue.length === 10)` so February 29 is not clamped to 28 when only 1 digit of the year has been typed (Fixes **REG-140**).
3	FRONTEND	`PinCodeField.jsx:167, 335-361`	Allow `cols.area` changes to re-trigger reverse lookup and prevent `lookup(foundPin, true)` from raising a false district mismatch when PIN was resolved via `state + area` fallback (Fixes **REG-064**).
4	FRONTEND	`RegistrationPreview.jsx:1924, 2850`	Capture `originalCombinedPropertiesRef.current` after initial form/child hydration settles so **Update** remains disabled until actual user input (Fixes **REG-110**).
5	FE + BE	`api.js:2416-2441` & `controller.ts:2314`	Check `if (response?.data?.statusCode >= 400)` in `api.js` , throw proper NestJS HTTP exceptions on conflict/error, and validate `mother_uhid` existence on Newborn save (Fixes **REG-090, REG-191**).
6	FE + CONFIG	`RegistrationPreview.jsx:2882` & Webpart Builder	Invoke `validateFieldValue()` on all populated fields inside `handleSave()` ; set `dateOption: "past_today"` on DOB, `minLength: 10` on WhatsApp/Emergency Mobile, and `isFirstName/Middle/Last` flags (Fixes **REG-033, REG-040, REG-050, REG-026, REG-027, REG-031**).

8. Complete Test Case Execution Ledger (Part 1 of 3: REG-001 to REG-035)

Cases 1–30 of 87

CASE ID	CATEGORY	STATUS	TEST CASE TITLE	ACTUAL OBSERVED RESULT	CREATED UHID
REG-001	happy	PASS	New Registration with only mandatory fields is created and persisted	created (UHID ASNR0000000445); toast: Registration Number: ASNR0000000445	`ASNR0000000445`
REG-002	happy	SKIPPED	New Registration with every optional field filled is created and persisted	disabled in case file	—
REG-003	happy	PASS	Female patient with title Mrs. is created with gender Female	created (UHID ASNR0000000404); toast: Registration Number: ASNR0000000404	`ASNR0000000404`
REG-004	dob-age	SKIPPED	Typing age 30 instead of DOB formats age and auto-calculates DOB	disabled in case file	—
REG-010	mandatory	PASS	Saving an empty form is blocked with a required-field message	blocked; toast: Please fill the required field: First Name	—
REG-011	mandatory	PASS	Missing First Name blocks save	blocked; toast: Please fill the required field: First Name; errors {"First Name":"This…	—
REG-012	mandatory	SKIPPED	Missing Last Name blocks save	disabled in case file	—
REG-013	mandatory	SKIPPED	Missing DOB (and age) blocks save	disabled in case file	—
REG-014	mandatory	PASS	Missing Mobile No blocks save	blocked; toast: Please fill the required field: Mobile No; errors {"Mobile No":"This f…	—
REG-015	mandatory	SKIPPED	Missing Address (marked * on screen) blocks save	disabled in case file	—
REG-016	mandatory	SKIPPED	Missing Pin Code (marked * on screen) blocks save	disabled in case file	—
REG-017	mandatory	SKIPPED	First Name of only spaces is treated as empty	disabled in case file	—
REG-018	mandatory	SKIPPED	Choosing 'None' for required Gender blocks save	disabled in case file	—
REG-019	mandatory	PASS	Choosing 'None' for required Title blocks save	blocked; toast: Please fill the required field: Title	—
REG-037	name	FAIL (GAP)	Title 'Mrs.' with Gender 'Male' is flagged as inconsistent	created (UHID ASNR0000000422); toast: Registration Number: ASNR0000000422	`ASNR0000000422`
REG-020	phone	PASS	9-digit mobile number shows an error and blocks save	blocked; toast: Please fill the required field: Pin Code (Permanent Address); errors {…	—
REG-021	phone	SKIPPED	Typing 11 digits into mobile keeps only 10	disabled in case file	—
REG-022	phone	SKIPPED	Mobile starting with 5 (not a valid Indian mobile) is rejected	disabled in case file	—
REG-023	phone	SKIPPED	Mobile with a leading 0 is accepted after dropping the 0	disabled in case file	—
REG-024	phone	SKIPPED	Letters typed into mobile are stripped	disabled in case file	—
REG-025	phone	FAIL (GAP)	All-same-digit mobile 9999999999 is rejected	created (UHID ASNR0000000423); toast: Registration Number: ASNR0000000423	`ASNR0000000423`
REG-026	phone	FAIL (GAP)	5-digit WhatsApp number is rejected	created (UHID ASNR0000000424); toast: Registration Number: ASNR0000000424	`ASNR0000000424`
REG-027	phone	FAIL (GAP)	3-digit emergency contact mobile is rejected	created (UHID ASNR0000000425); toast: Registration Number: ASNR0000000425	`ASNR0000000425`
REG-030	name	FAIL (GAP)	Digits in First Name are rejected	created (UHID ASNR0000000426); toast: Registration Number: ASNR0000000426	`ASNR0000000426`
REG-031	name	FAIL (GAP)	Special characters in First Name are rejected	created (UHID ASNR0000000427); toast: Registration Number: ASNR0000000427	`ASNR0000000427`
REG-032	name	SKIPPED	120-character First Name is limited or rejected	disabled in case file	—
REG-033	name	FAIL (BUG)	Full name typed into First Name is split into middle and last name	values {"first_name":"Agent Testing UserEighty","middle_name":"","last_name":""}	—
REG-034	name	SKIPPED	Hindi (Unicode) First Name is accepted	disabled in case file	—
REG-035	name	SKIPPED	Leading/trailing spaces in names are trimmed when saved	disabled in case file	—
REG-040	dob-age	FAIL (GAP)	Future DOB is rejected	created (UHID ASNR0000000428); toast: Registration Number: ASNR0000000428	`ASNR0000000428`

9. Complete Test Case Execution Ledger (Part 2 of 3: REG-040 to REG-103)

Cases 31–60 of 87

CASE ID	CATEGORY	STATUS	TEST CASE TITLE	ACTUAL OBSERVED RESULT	CREATED UHID
REG-041	dob-age	SKIPPED	Impossible date 31/02/2000 is not accepted as typed	disabled in case file	—
REG-042	dob-age	SKIPPED	Day 45 is capped to 31 by the DOB mask	disabled in case file	—
REG-043	dob-age	SKIPPED	DOB today gives age 0Y 0M 0D	disabled in case file	—
REG-044	dob-age	FAIL (GAP)	DOB in 1900 (age 126) is rejected	created (UHID ASNR0000000429); toast: Registration Number: ASNR0000000429	ASNR0000000429
REG-045	dob-age	FAIL (GAP)	Age 150 is rejected	created (UHID ASNR0000000430); toast: Registration Number: ASNR0000000430	ASNR0000000430
REG-050	email	FAIL (GAP)	Invalid email 'agent@' is rejected	created (UHID ASNR0000000431); toast: Registration Number: ASNR0000000431	ASNR0000000431
REG-060	address	PASS	PIN 110001 fills State/District/City and asks to choose an Area	values {"country":"India","state":"DELHI","district":"NEW DELHI","city":"NEW DELHI","p…	—
REG-061	address	PASS	Unknown PIN 000000 shows 'not recognized'	values {"state":"","pin_code":"000000","hint:pin_code":"PIN code not recognized — plea…	—
REG-062	address	PASS	5-digit PIN does not trigger a lookup	values {"state":"","pin_code":"11000"}	—
REG-063	address	SKIPPED	Letters typed into PIN are stripped	disabled in case file	—
REG-064	address	FAIL (BUG)	Choosing State, District and Area without a PIN fills the PIN (reverse lookup)	values {"pin_code":"","state":"DELHI","district":"NEW DELHI","area":"F F C Okhla"}	—
REG-065	address	SKIPPED	PIN 246149 fills UTTARAKHAND / PAURI GARHWAL	disabled in case file	—
REG-066	address	PASS	Changing State after a typed PIN clears District/City, keeps the typed PIN and warns of the mismatch	values {"district":"","city":"","pin_code":"110001","state":"UTTARAKHAND","hint:pin_co…	—
REG-067	address	SKIPPED	'Mark current address as permanent' patient is created	disabled in case file	—
REG-069	address	PASS	Filling only Current Address (Permanent tab untouched) blocks save on Permanent Pin Code	blocked; toast: Please fill the required field: Pin Code (Permanent Address)	—
REG-068	address	OBSERVED	Very long (600 char) address is accepted or limited	values {"address":"Agent test address block, Agent test address block, Agent test addr…	—
REG-070	payer	PASS	Default payer is Self-Paid with Payer and Sponsor = Cash	values {"payer_type":"Self-Paid","payer":"Cash","sponsor":"Cash"}	—
REG-080	registration-type	SKIPPED	New Born Registration without Mother UHID (marked *) is blocked	disabled in case file	—
REG-081	registration-type	SKIPPED	International Registration with mandatory fields is created	disabled in case file	—
REG-082	registration-type	PASS	Outside Registration with mandatory fields is created	created (UHID ASOR240000000012); toast: Registration Number: ASOR240000000012	ASOR240000000012
REG-083	registration-type	PASS	Diagnostic Registration with mandatory fields is created	created (UHID ASDR240000000001); toast: Registration Number: ASDR240000000001	ASDR240000000001
REG-084	registration-type	PASS	Staff Registration without Staff Details (marked *) is blocked	blocked; toast: Please fill the required field: Staff Details	—
REG-085	ui	PASS	Switching registration type does not show 'required' errors before Save	values {}	—
REG-090	duplicate	PASS	Registering the exact same patient twice is not reported as a success	duplicate-warning	ASNR0000000434
REG-100	search	PASS	Search 'Agent Testing' finds the agent-created patients	found (32 records found)	—
REG-101	search	FAIL (BUG)	Search 'Vineet' finds the existing patient Vineet	not-found (0 records found)	—
REG-102	search	PASS	Search 'Vivek' finds the existing patient Vivek	found (23 records found)	—
REG-103	search	PASS	Search by the mobile number of REG-001 finds it	found (opened single match)	—
REG-104	search	FAIL (BUG)	Search with a regex character '(' does not error	error (0 records found)	—
REG-105	search	PASS	Search for a nonsense name returns 0 records	not-found (0 records found)	—

10. Complete Test Case Execution Ledger (Part 3 of 3: REG-104 to REG-220)

Cases 61–87 of 87

CASE ID	CATEGORY	STATUS	TEST CASE TITLE	ACTUAL OBSERVED RESULT	CREATED UHID
REG-110	update	FAIL (BUG)	Update button stays disabled until something changes	field-state; Update button enabled before changes	`ASNR0000000445`
REG-111	update	PASS	Editing the email of REG-001 and submitting remarks updates the record	updated; Update button enabled before changes	`ASNR0000000433`
REG-130	real-names	PASS	Apostrophe surname D'Souza is saved exactly	created (UHID ASNR0000000446); toast: Registration Number: ASNR0000000446	`ASNR0000000446`
REG-131	real-names	PASS	Hyphenated first name Anne-Marie is saved exactly	created (UHID ASNR0000000436); toast: Registration Number: ASNR0000000436	`ASNR0000000436`
REG-132	real-names	PASS	Initial with a dot 'S.' as first name is saved exactly	created (UHID ASNR0000000447); toast: Registration Number: ASNR0000000447	`ASNR0000000447`
REG-133	real-names	PASS	Single-letter first name 'A' is accepted	created (UHID ASNR0000000448); toast: Registration Number: ASNR0000000448	`ASNR0000000448`
REG-134	real-names	PASS	What 'Agent123' is actually saved as (digit stripping)	created (UHID ASNR0000000439); toast: Registration Number: ASNR0000000439	`ASNR0000000439`
REG-140	dob-age	FAIL (BUG)	Leap day 29/02/2000 is accepted	values {"dob":"28/02/2000","age":"26Y 7M 30D"}	—
REG-141	dob-age	OBSERVED	29/02/2001 (not a leap year) is not accepted as typed	values {"dob":"28/02/2001"}	—
REG-142	dob-age	PASS	Typing age then DOB recalculates age from DOB	values {"age":"36Y 2M 14D","dob":"15/08/1990"}	—
REG-150	email	PASS	Email with plus-addressing is accepted and saved	created (UHID ASNR0000000449); toast: Registration Number: ASNR0000000449	`ASNR0000000449`
REG-151	email	FAIL (GAP)	Email with spaces is rejected	created (UHID ASNR0000000450); toast: Registration Number: ASNR0000000450	`ASNR0000000450`
REG-160	phone	FAIL (GAP)	Emergency contact mobile equal to patient mobile is flagged	created (UHID ASNR0000000442); toast: Registration Number: ASNR0000000442	`ASNR0000000442`
REG-161	phone	PASS	Country Australia re-validates a 10-digit Indian mobile	blocked; toast: Please fill the required field: Pin Code (Current Address)	—
REG-170	payer	PASS	Payer Type Corporate without choosing a Payer blocks save	blocked; toast: Please fill the required field: Payer	—
REG-171	payer	PASS	Payer Type Insurance without choosing a Payer blocks save	blocked; toast: Please fill the required field: Payer	—
REG-172	payer	PASS	Corporate payer with first Payer and Sponsor is created	created (UHID ASNR0000000443); toast: Registration Number: ASNR0000000443	`ASNR0000000443`
REG-180	identity	PASS	Id Type Foreigner without an Id Number is blocked	blocked; toast: Please fill the required field: Passport Issue Place	—
REG-190	registration-type	PASS	New Born with Mother UHID of an existing patient is created	created (UHID ASNBR240000000007); toast: Registration Number: ASNBR240000000007	`ASNBR240000000007`
REG-191	registration-type	FAIL (GAP)	New Born with a Mother UHID that does not exist is rejected	created (UHID ASNBR240000000008); toast: Registration Number: ASNBR240000000008	`ASNBR240000000008`
REG-192	registration-type	FAIL (GAP)	New Born with DOB 30 years ago is rejected	created (UHID ASNBR240000000009); toast: Registration Number: ASNBR240000000009	`ASNBR240000000009`
REG-200	data-integrity	PASS	Double-clicking Save creates exactly one patient	1 create request(s), 1 record(s): ASNR0000000444	`ASNR0000000444`
REG-210	search	PASS	Search by the UHID of REG-001 opens it	found (opened single match)	—
REG-211	search	PASS	Search in lower case 'agent testing' finds agent patients	found (41 records found)	—
REG-212	search	PASS	Search with surrounding spaces ' Agent ' still finds patients	found (42 records found)	—
REG-220	update	PASS	Changing the mobile of REG-001 to 9 digits blocks the update	blocked; Update button enabled before changes; toast: Mobile No: Enter a valid 10-digi…	`ASNR0000000445`
REG-120	identity	SKIPPED	Upload ID proof PDF attaches it to the patient	Pending: Upload ID Proof modal locators (run discovery first).	—

Note on `REG-101` ("Search 'Vineet' finds the existing patient Vineet"): Both the walkthrough video (`pr_sec_010.png`) and live automated test runs returned `"0 records found"` for `"Vineet"`, whereas `"Vivek"` (`REG-102`) returned 23 records and `"Agent Testing"` (`REG-100`) returned 42 records. This confirms the top-bar search works for normal strings and that no active patient named `"Vineet"` is currently seeded in the `Miracle HIS Demo` tenant.